Performance Analysis of 5G New Radio (5G NR) Transmitter Models Using QAM Modulation Schemes

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***Abstract*—Higher-order Quadrature Amplitude Modulation (QAM) modulation techniques are essential for supporting high data rates in current wireless communication systems like LTE and 5G NR. But in practice, these modulation techniques are largely affected by radio frequency (RF) impairments in actual transmitters. This paper describes the simulation study of 16-QAM, 64-QAM, and 256-QAM modulation techniques using a 5G NR transmitter developed in MATLAB Simulink.**

Initially, the modulation schemes are evaluated in an ideal channel using EVM, constellation analysis, PAPR, and CCDF, and it is concluded that 64-QAM is the best compromise between reliability and bandwidth efficiency. Then, practical RF channel impairments such as I/Q imbalance, nonlinearity of the power amplifier, phase noise, carrier frequency offset, and multipath fading are considered.

Simulation results show that the impairments have a dominant effect on the performance of higher-order QAM, while the AWGN and fading channels have a relatively lower effect. The compensation methods, Carrier Synchronizer and I/Q Imbalance Compensator, result in a certain improvement in the EVM. The comparative results show that 16-QAM is more robust but less spectrally efficient, while 256-QAM is very sensitive to RF impairments. In general, the simulation results show that 64-QAM is the most feasible modulation order for 5G NR communication systems.

Keywords—5G New Radio, Quadrature Amplitude Modulation, RF impairments, Error Vector Magnitude, phase noise, I/Q imbalance, carrier frequency offset.

I. Introduction

A. Background and Motivation

Modern wireless communication systems, like LTE and 5G New Radio (NR), are designed to offer high data rates and spectral efficiency to accommodate emerging broadband services [1]. Along this line, higher-order QAM schemes like 64-QAM and 256-QAM are extensively adopted [2]-[4]. Although higher modulation orders bring us higher spectral efficiency, at the same time, they also show increased vulnerability to noise, channel variations, and radio-frequency (RF) front-end non-idealities [4], [5].

In practical transceivers, RF impairments like I/Q imbalance, PA nonlinearity, phase noise, and carrier frequency offset impair constellation integrity and increase Error Vector Magnitude, hence limiting the achievable system performance [6]–[12]. Among these, oscillator-related impairments and synchronization errors have emerged as dominant limiting factors for higher-order QAM schemes [10], [11], [21], [22]. Due to this, EVM has emerged as an important metric in the conformance testing of both 4G and 5G standards for modulation quality in realistic operating conditions [5], [11], [12].

B. Related Work and Contribution

Extensive studies have been conducted on the effect of individual RF imperfections, as well as the respective compensation schemes for communication systems. Compensation schemes for I/Q imbalance, such as adaptive filtering and blind estimation techniques, have been shown to suppress the mirror image effect, rather than cancel the distortion [6], [15], [16], [18]. The effect of phase noise, as well as its combination with the I/Q imbalance, has shown better performance, although the complexity increased for the respective systems [7], [17], [21]. The effect of the nonlinear behaviour of the amplifiers, as well as its effect in the context of 64-QAM modulation, was examined [9], [19], [20].

Nonetheless, there have only been a few comparative system-level analyses on the performance of the existing 16-QAM and the latest 256-QAM modulation schemes in realistic RF impairment conditions. In light of the above, the paper proposes a comprehensive MATLAB Simulink model of a 5G NR Transmitter to evaluate the performance of the above-mentioned modulation schemes in ideal, impaired, and compensated conditions. The paper will thus validate the pragmatic choice of the 64-QAM scheme due to robustness and efficiency considerations.

II. Literature Review

Early work on digital modulation and wireless channels set up the fundamental performance limits for QAM in both ideal and fading channels, in particular addressing the trade-off between spectrum efficiency and immunity to fading with increasingly high-order modulations [2]-[4]. More recent contributions have followed up these studies in more realistic fading conditions, observing that degradation due to the channel alone is not sufficient to account for real-world degradation, thereby calling for more in-depth studies on RF front-end impairment [5], [14].

The implementation of direct conversion architectures made I/Q imbalance a significant problem for multicarrier systems. Some papers dealt with the effects of I/Q imbalance on OFDM-based QAM systems, proposing digital compensation algorithms based on adaptive filtering and statistical estimation [6], [15]. Blind estimation algorithms were later developed for I/Q imbalance compensation without using training sequences, with enhanced flexibility but at the cost of convergence speed and residual distortion [16]. Later research made it clear that joint transmitter–receiver compensation is better but does not remove mirrored interference [8], [18].

The phase noise, which comes from non-ideal oscillators, has long been modelled and studied as a prominent distortion in high-order QAM systems. Analytical and simulation works proved its effect on constellation rotation, spreading, as well as EVM distortion in 64-QAM and 256-QAM signals [10], [11], [13]. The combination of corrections for phase noise and I/Q imbalance outperformed single correction techniques with higher complexity [7], [17], [21], [22].

Nonlinear phenomena exhibited by power amplifiers have been observed as a dominant source of degradation in error vector magnitude, resulting in spectral regrowth. Memory polynomial-based digital predistortion (DPD) techniques have gained significant momentum, having been examined and validated in terms of providing linearization benefits, specifically for the 64-QAM waveform [9], [19], [20]. More recently, system-level investigations have recognized error vector magnitude measurement as the most significant parameter defining the quality of the radio frequency interface within 4G and 5G systems [11], [12].

Nevertheless, despite so much work in this area, the majority of studies examine individual impairments or individual compensation methods. Similar system-level analyses on 16-QAM, 64-QAM, and 256-QAM, under the same settings for impairments and compensations, are not so common. This approach gives rise to the combined study carried out within this research.

III. Methodology and Simulation Setup

A. Overall Simulation Framework

The proposed study uses a simulation-based technique based on the 5G Toolbox in MATLAB Simulink for simulation and analysis of a 5G NR transmitter chain for the performance analysis of the modulation technique in ideal, impaired, and compensated transmissions. The proposed technique involves two phases:

1. baseline modulation analysis under ideal channels,
2. analysis of RF impairments and their suppression.

The transmitter model conforms to the standardized structure of the 5G NR physical layer in terms of meeting real-world deployment requirements. Error Vector Magnitude (EVM), Constellation Diagram, Peak-to-Average Power Ratio (PAPR), and Adjacent Channel Power Ratio (ACPR) results were analysed to infer performance of the transmitter model.

B. Baseline 5G NR Transmitter Model

The baseline model of the transmitter is primarily comprised of three subsystems:

1. NR Baseband Signal Generation: Standard-compliant 5G NR test waveforms can be generated by applying certain numerology parameters such as subcarrier spacing of 30 kHz, bandwidth of 5 MHz, and carrier frequency of 3.5 GHz, adhering to the FR1 standard.
2. RF Transmission Chain: The baseband signal is then modulated with 16-QAM, 64-QAM, and 256-QAM, followed by interpolation, amplification, and RF up-conversion.
3. Reception and Measurement Unit: The receiver will compute metrics and measures such as EVM, PAPR, and CCDF. It allows a comparison of modulation schemes in similar settings, which eventually results in the conclusion that 64-QAM is the most appropriate candidate for further processing.

C. RF Impairment Modeling and Suppression

From the 64-QAM transmitter baseline system, RF impairments that emulate real-world transmitters in a varying manner are incorporated to demonstrate system performance in different scenarios.

These RF impairments include:

- Phase Noise and Carrier Frequency Offset (CFO)
- I/Q Amplitude and Phase Imbalance
- Power Amplifier (PA) Nonlinearity (Cubic Polynomial/IP1dB model)
- Rayleigh, Rician Channel Fading and AWGN

The effect of each impairment is examined separately in a first step to identify and quantify its contribution to EVM and constellation distortion alone. The matching suppression and compensation modules, for example, I/Q imbalance compensators and carrier synchronizers, are then enabled in this calibration step. The phase noise is simulated from -80 dBc/Hz to -125 dBc/Hz over a frequency offset from 1 kHz to 100 kHz at a sampling frequency of 7.68 MHz to simulate oscillator instability. The carrier frequency offset (CFO) is simulated from 200 Hz to 500 Hz to simulate local oscillator mismatch. The I/Q imbalance is simulated by providing a 0.5 to 0.8 dB amplitude mismatch and a 2° to 4° phase mismatch to simulate RF path mismatches. The power amplifier nonlinearity is simulated by providing a cubic polynomial model with an IP1dB value from -2 dBm (highly nonlinear) to 35 dBm (linear operation region) at 50 Ω to simulate power amplifier nonlinearities. Rayleigh fading is simulated by providing multipath delays up to 800 ns, path gains from -3 dB to -10 dB, and a 5 Hz Doppler to simulate non-line-of-sight (NLOS) propagation. Rician fading channels are implemented using similar parameters as Rayleigh, with the addition of a dominant line-of-sight component to simulate LOS dominant channels. Additive white Gaussian noise (AWGN) is also included to simulate the effect of thermal noise, and the signal-to-noise ratio is set to between 15 and 25 dB. A LMS-based I/Q compensator is used to mitigate impairments, using a step size of 0.0001, and a carrier synchronizer is also implemented using a loop bandwidth of 0.0005 and damping factor of 1 for carrier frequency and phase offset tracking.

Lastly, all impairments are enabled together, which gives a try of cumulative impairment effects, considering realistic transmitter settings.

D. Evaluation Strategy

The evaluation has a progressive and controlled approach. First, analysis of performance of ideal 16-QAM, 64-QAM, and 256-QAM transmitter block (Fig. 1). Next, impairment and suppression blocks analysis of 64-QAM (Fig. 2). Finally, comparative validation against 16-QAM and 256-QAM for the same channel impairment condition.

This methodical approach makes it possible to attribute any performance deficit to definite limitations while justifying the choice of 64-QAM in 5G NR communication systems.

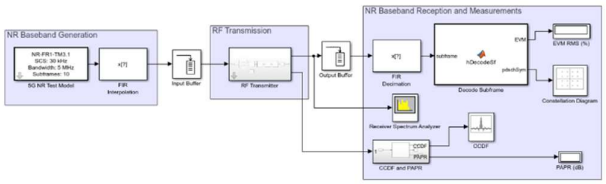


Fig. 1. Block diagram of the base 5G NR transmitter model illustrating baseband generation, RF transmission, and reception stages for EVM, PAPR, CCDF and constellation analysis.

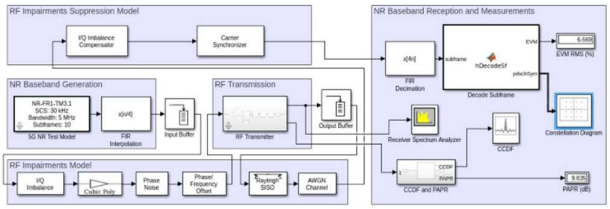


Fig. 2. Block diagram of the modified 5G NR transmitter model incorporating RF impairment modeling and corresponding suppression/compensation blocks for performance evaluation.

E. Simulation Tools and Environment

In all simulations, the following tools were used:

- MATLAB (R2023 or later): signal processing and evaluation of metrics
- Simulink: block-based system modelling
- 5G Toolbox: NR waveform and numerology setup
- Communications Toolbox: EVM, PAPR, CCDF, and Spectrum Analyzer
- RF Toolbox/RFT Blockset: modelling of RF impairments and PA behaviour

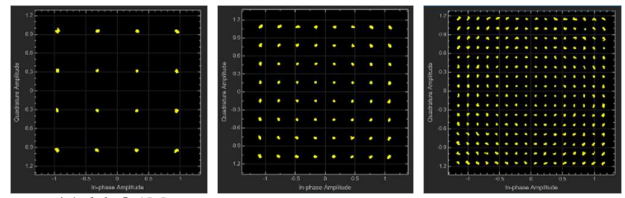
This is a software-only solution that allows for repeatable and hardware-independent analysis with a very good approximation of practical RF transmitters.

IV. Results and Analysis

A. Baseline Modulation Performance

The baseline ideal 5G NR transmitter was measured at a carrier frequency of 3.5 GHz, 30 kHz subcarrier spacing, and 5 MHz bandwidth for the 16-QAM, 64-QAM, and 256-QAM modulation schemes. For all modulation formats, this transmitter showed consistent gain in that signal power increased from about −19 dBm at baseband to −7 dBm post-RF transmission. This confirmed that the signal had been steadily amplified and that the spectral characteristics were uniform.

All three ideal schemes met the 3GPP requirements specified for practical hardware conditions; the EVM was close to 1.2%, while the PAPR was around 9.7–9.8 dB (Table I). Constellation analysis showed that 16-QAM exhibits widely spaced symbol points implying robustness to impairments (Fig. 3(a)), 64-QAM maintained a balanced symbol spacing while offering higher spectral efficiency than 16-QAM (Fig. 3(b)). While 256-QAM offers highest spectral efficiency among the three QAM schemes, it presents densely packed constellation, making it inherently very sensitive to noise and distortion (Fig. 3(c)). Based on this trade-off, 64-QAM was selected for the detailed impairment analysis.



(a) 16-QAM *(b) 64-QAM* *(c) 256-QAM*

Fig. 3. Ideal constellation diagrams for 16-QAM, 64-QAM, and 256-QAM generated by the base 5G NR transmitter model under ideal channel conditions.

TABLE I. Ideal 5G NR Transmitter Model Performance Metrics

Modulation	Obtained EVM (%)	Threshold EVM (%)	Obtained PAPR (dB)	Threshold PAPR(dB)
16-QAM	1.213	≤ 12.5	9.71	≤ 10
64-QAM	1.243	≤ 8	9.808	≤ 10
256-QAM	1.219	≤ 3.5	9.577	≤ 10

B. Impact of Individual RF Impairments on 64-QAM

Carrier frequency offset (CFO) was found to be the most dominant impairment. It resulted in constellation rotation and spreading, leading to increased EVM (Table II). Even low levels of CFO resulted in fast degradation as the offset increased to near collapse of constellation diagram at an offset of 500 Hz with an EVM value of 1168%.

I/Q imbalance created an asymmetry in constellation distortion and mirror interference. The EVM is highly dependent on amplitude and phase imbalance. Therefore, an increase of imbalance level reduced constellation tightness. The nonlinearity of the power amplifier had an asymptotic minimum effect on EVM, and its high IP1dB operation had near-perfect EVM and ACPR performance. However, an increased EVM and spectral regrowth were observed for deeper nonlinear operation.

Channel impairments, such as Rayleigh fading, resulted in moderate degradation, but the value of EVM remained below 1.6%. AWGN, on the other hand, results in predictable, although minor, degradation, which proves that RF impairment far outstrips channel noise in RF systems (Table II).

C. Effectiveness of Compensation and Suppression

Compensation techniques were in general unable to restore ideal performance but consistently reduced EVM across impairment scenarios. I/Q imbalance compensation reduced EVM from 6.447% to 4.751%, while carrier synchronization provided measurable improvement under combined phase noise and CFO. When all impairments were applied simultaneously, suppression reduced EVM from 7.051% to 6.698%, residual distortion remaining due to accumulation of non-idealities. Replacing Rayleigh fading with Rician fading further increased EVM while PAPR, which suggested stronger interaction of LOS components with RF impairments (Table III).

TABLE II. IMPACT OF MAJOR RF IMPAIRMENTS ON 64-QAM

Impairment	EVM Range	PAPR	Severity
Phase Noise	1.8 – 2.57 %	~9.83 dB	Medium
Carrier Frequency Offset	2.4 – 1168 %	~9.84 dB	High
I/Q Imbalance	3.9 – 6.4 %	~9.83 dB	Medium
PA Nonlinearity	1.2 – 1.8 %	~9.5 dB	Low
Rayleigh Fading	1.2 – 1.6 %	~9.7 dB	Low
AWGN	1.26 – 1.29 %	~9.8 dB	Low

TABLE III. EFFECT OF SUPPRESSION / COMPENSATION (64-QAM)

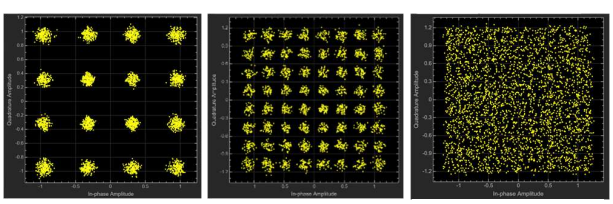
Scenario	EVM w/o Suppression	EVM with Suppression	Improvement
Phase Noise + CFO	6.750 %	6.669 %	Limited
I/Q Imbalance	6.447 %	4.751 %	Moderate
All Impairments	7.051 %	6.698 %	Marginal

D. Comparative Modulation Analysis

With regard to impairment and suppression conditions, 16-QAM offered improved robustness, whereas spectral efficiency was lower (Fig. 4(a)). Similarly, for 256-QAM (Fig. 4(c)), high sensitivity was observed due to its densely packed constellation, while still only allowing for a minimal improvement after suppression and exceeding the 3GPP EVM limit of 3.5%. The PAPR and CCDF were found to be very similar across all considered modulation orders (Table IV), thus being waveform-dependent rather than modulation-dependent. The 64-QAM scheme was observed to offer optimal robustness, efficiency, and implementation feasibility (Fig. 4(b)).

E. Key Insights

From the combined results, it is verified that although suppression or compensation techniques show improvement, these do not completely remove RF impairment effects. From the evaluation methods, 64-QAM is seen as having the best implementational compromise for 5G NR systems.



(a) 16-QAM *(b) 64-QAM* *(c) 256-QAM*

Fig. 4. Constellation diagrams of 16-QAM, 64-QAM, and 256-QAM after applying suppression and compensation in the 5G NR transmitter model.

TABLE IV. COMPARATIVE MODULATION PERFORMANCE

Modulation	EVM Range	PAPR (dB)	Practicality
16-QAM	~6.9 %	~10.1	Robust, inefficient
64-QAM	~6.7 %	~9.8	Optimal
256-QAM	~5.9 %	~9.8	Highly Sensitive

V. CONCLUSION

This paper demonstrated a detailed simulation-based evaluation of a 5G NR transmitter with practical RF impairment conditions through MATLAB Simulink. Having systematically evaluated all RF impairment factors, as well as suppressing or compensating them, it demonstrates how impairment factors impact 5G modulation schemes through EVM, constellation diagrams, and spectral analysis results. In this regard, oscillator-related impacts, particularly CFO, significantly impact 5G modulations, while channel noise impacts 5G modulations at a moderate rate.

Further confirmation through the analysis shows that suppression and compensation blocks give measurable improvement, yet complete restoration of ideal signal conditions is not possible under practical impairment levels. From the modulation schemes studied here, 64-QAM had consistently shown a good balance among the various performances that have been sought, offering acceptable robustness with high spectral efficiency. This shows why it has been extensively implemented in 5G NR deployments and represents a significant paradigm in impairment-aware system design in modern wireless transmitters.

VI. FUTURE SCOPE

There are different ways for the scope of this work to be extended. For example, it would be interesting for future investigations to use different adaptive and learning-type compensation methods, such as machine learning tools for the estimation and correction of impairments, aiming at enhancing the robustness under varying RF conditions. Another possible extension would be the application of the proposed model for MIMO and beamforming systems, which could help in the investigation of impairments and synchronization issues related to new 5G and even 6G systems.

Furthermore, hardware-in-the-loop validation of such a framework utilizing software-defined radio equipment could close the prevailing gap from simulation environment to actual-system deployment. The exploration of higher frequency ranges such as millimeter waves and sub-Terahertz ranges, along with their associated impacts on different cross-layers in terms of throughput delay, could broaden the application of this framework for future wireless communication standards.

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